

An Exploratory Analysis:

Digital Infrastructure and Preventative Healthcare Utilization

SIADS 593: Milestone 1

Group 21

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Background, Motivation, & Intent

Our project examines the relationship between state-level digital infrastructure and healthcare engagement patterns across the United States.

Healthcare systems continue to advance and rely on digital tools, including online appointment scheduling through patient portals and telehealth services. Access to the internet and digital devices can create barriers to healthcare participation. A digital divide built from these factors could contribute to the existing healthcare disparities we see in the United States, especially in rural and low-income areas.

According to the National Healthcare Quality and Disparities Report, factors that influence healthcare disparities include social, economic, environmental, access, health literacy, and communication. These disparities are not new and reflect long standing structural and systematic inequalities rooted in contemporary and historical racism and discrimination.

Better internet access means better health. Expanding digital access can help more people get preventative care and reduce gaps in communities.

Our primary research question is:

[1] "Do states with better digital infrastructure show higher rates of preventive healthcare utilization and healthcare engagement?"

We want to identify potential correlations between digital connectivity indicators (internet access, device ownership, digital literacy) and healthcare utilization patterns (preventive care visits, health screenings, time since last doctor visit). Additionally, we seek to understand [2] whether individuals living in urban areas have greater internet access, and improved healthcare access, compared to those in rural areas.

Our analysis will compare state-level data from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) with the National Telecommunications and Information Administration's (NTIA) Internet Use Survey data. While we cannot establish causation, we can identify correlations that shed light on how digital infrastructure improvements might support better healthcare access and inform policy discussions about digital equity. Building on prior work that has highlighted rural digital health challenges, our project contributes by combining these datasets to explore how digital access both supports and reshapes preventive care. These insights are especially timely as policymakers grapple with ensuring that telehealth expansion reduces rather than widens disparities.



Data Sources



CDC's Behavioral Risk Factor Surveillance System (BRFSS)



BRFSS is the largest continuously conducted telephone-based health survey system in the U.S., collecting survey-based health and behavior data from 1989-2023. This includes risk behaviors like smoking and drinking to avenues like obesity, exercise, sleep, and preventive care. It is available by state and includes demographics such as age, gender, income, education, employment, and race.



The datasets for BRFSS are large, ~1GB per year of data. For this analysis, we used years 2019, 2021, & 2023. Each year's dataset contains approximately 425,000 records.



Format: XPT (SAS Transport File)
2019: 418,268 rows x 342 columns
2021: 438,693 rows x 303 columns
2023: 433,323 rows x 350 columns



NTIA's Internet Use Survey



NTIA's Internet Use Survey provides information about household access to computers and the use of the Internet from 1994-2023. It measures each state's population internet use, device use, household connectivity, and online activities.



This dataset contains data on all years collected from 1994-2023, ~2200 KB, and has 835 records.



Format: CSV
1994-2023: 835 rows x 348 columns

Data Cleaning & Manipulation - BRFSS

1. **Original files & Loading:** BRFSS datasets in .XPT format (SAS Transport). Shape: 2019: (418,268, 342), 2021: (438,693, 303), 2023: (433,323, 350). Available through Google Drive in .README. Loaded using `load_brfss_year(year)` with `pd.read_sas()`
2. **Standardize Variable Names:** BRFSS renames variables across years (e.g., *PERSDOC2* → *PERSDOC3*, *CHOLCHK2* → *CHOLCHK3*). Used mapping dictionaries and codebook verification for consistency. Ensured uniform definitions across years (e.g., checkups = past year, doctor access = ≥1 provider, cholesterol check = past 5 years).
3. **Extract Variables of Interest:** Reduced from 342+ columns → 10 variables (5 healthcare + 3 demographic + 1 ID + 1 weight) = (418,268, 342) → (418,268, 11). **Note:** Demographic controls (*_METSTAT*, *_EDUCAG*, *_INCOMG1*) extracted but not included in final state-level analysis.
4. **Handle Missing Values:** Used `cleaning_utils.clean_brfss_missing_values()` to convert BRFSS missing codes (7="Don't know", 9="Refused", 77, 99) to NaN (~200,000 per year). Dropped records with missing *_STATE* values using `dropna(subset=['_STATE'])` (lost ~2% of records per year)
5. **Aggregate to State Level with Survey Weights:** Applied *_LLCPWT* to calculate weighted state percentages. Formula: $\text{State \%} = (\text{Sum of weights where response} = \text{"Yes"}) / (\text{Sum of weights for all valid responses}) \times 100$. Weights correct for rural oversampling. Used `aggregate_to_state(df, year)` function → (409,105, 11) → (51, 7) per year
6. **Combine Years:** Concatenated 3 years: `brfss_state_all_years = pd.concat([brfss_2019_state, brfss_2021_state, brfss_2023_state])`. Final shape: (154, 7) covering 51+52+51 state observations across 2019, 2021, 2023

Healthcare Engagement (% of adults)	Description
CHECKUP1_pct	Had routine checkup in past year
PERSDOC3_pct	Has personal healthcare provider
FLUSHOT7_pct	Received flu vaccination
EXERANY2_pct	Participated in physical activity (past month)
CHOLCHK3_pct	Had cholesterol screening (past 5 years)

***Plus:** *_STATE* (FIPS code), *state_abbr* (added for merging), *YEAR* (2019, 2021, 2023)

Data Cleaning & Manipulation - NTIA

1. **Original file:** The NTIA dataset is in .CSV format with unusual structure: rows represent survey variables, columns represent states. Shape: (835, 348). Available for download through Google Drive in the .README. Loaded using `load_ntia_data()` with `pd.read_csv()`
2. **Filter Variables & Years:** Selected 7 digital infrastructure variables per year from 835 available, filtered to November 2019, 2021, 2023 to match BRFSS timeframes. Result: (835, 348) → (21, 348)
3. **Select Universe for Variable Duplicates:** Some variables measured across multiple populations (e.g., wiredHighSpeedAtHome as % of internet users vs % of households). Selected one consistent universe per variable using `universe_selection` dictionary. Household-level (isHouseholder) for infrastructure, adult-level (isAdult) for devices.
4. **Reshape Data Structure:**
 - a. State data spread across columns (ALProp, AKProp...), needed rows to represent state-variable-year combinations
 - b. Used `ntia_reshape.melt()` to convert state columns into rows → (1,092, 4)
 - c. Removed national averages (`state_abbr != 'us'`) → (1,071, 4)
 - d. Used `pivot_table()` to convert variables into separate columns → (153, 9)

Digital Infrastructure Variable	Description + Universe
homeInternetUser	Proportion of adults using internet at home + isAdult
emailUser	Proportion of adults using email + adultInternetUser
callConfUser	Proportion with video/voice calling capability (telehealth readiness) + adultInternetUser
pcOrTabletUser	Proportion with computer or tablet (patient portal access) + isAdult
mobilePhoneUser	Proportion with smartphone + isAdult
wiredHighSpeedAtHome	Proportion of households with wired broadband + isHouseholder
tooExpensiveMainReason	Proportion citing cost as main barrier to internet + isHouseholder

Merging BRFSS & NTIA

1. **State Identifier Challenge:** BRFSS uses numeric FIPS codes (_STATE: 1, 2, 4...) while NTIA uses abbreviations (state_abbr: AL, AK, AZ...)
2. **Solution:** Created `state_mapping` dictionary (1→AL, 2→AK...) and added `state_abbr` column to BRFSS:
`brfss_state_all_years['state_abbr'] = brfss_state_all_years['_STATE'].map(state_mapping)`
3. **Inner Merge:** Joined on `state_abbr + YEAR`: `merged_data = brfss_state_all_years.merge(ntia_wide, on=['state_abbr', 'YEAR'], how='inner')`
4. **Final Result:** (146, 15) - Lost 8 territories present in BRFSS but not NTIA. Final dataset covers 50 states × 3 years with 5 healthcare + 7 digital infrastructure metrics. Saved as `merged_brfss_ntia.csv`

Dataset	Rows Before Merge	Rows After Merge	Data Loss
BRFSS (state-level)	154	146	-
NTIA (state-level)	153	146	-
Final Merged	-	146	8 territories

Lost observations: Puerto Rico, Guam, Virgin Islands (present in BRFSS but not NTIA)

Exploratory Data Analysis - Distributions & Outliers

Summary Statistics Overview

Healthcare Engagement

(state-level percentages, n=146)

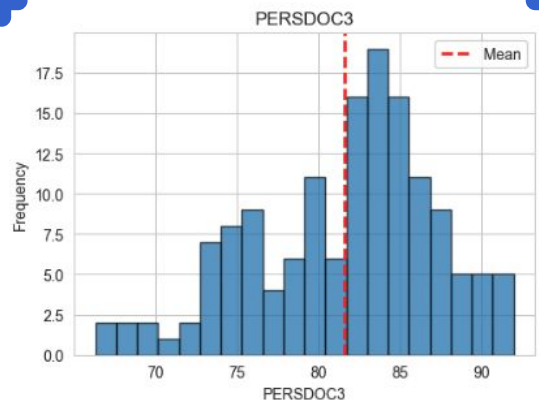
- **Checkups:** Avg ~77% – generally high, but some 2021 dips (e.g., CA, NM).
- **Personal Doctor:** Avg ~82% – widest variation (66–92%).
- **Flu Shots:** Avg ~44% – moderate, large cross-state spread.
- **Exercise:** Avg ~75% – fairly consistent across states.
- **Cholesterol Checks:** Avg ~84% – tight clustering, slight skew.

Digital Infrastructure

(state-level proportions, n=146):

- **Email Use:** ~92% – most consistent
- **Video Calling:** ~60% – largest state differences.
- **Home Internet:** ~79%.
- **Broadband:** ~71%.

Distribution Analysis



State-level health indicators reveal particular patterns across different metrics. Access to a personal doctor (PERSDOC3) is widespread (66–92%) but skewed, with some states clustering high while others lag. This could reflect differences in insurance coverage, Medicaid expansion, and the availability of primary care. However, both routine checkups and exercise demonstrate more consistent engagement, with bell-shaped distributions showing most states in a mid-to-high band. Flu shot rates are more evenly scattered, showing large cross-state differences, while cholesterol checks are broadly adopted yet unevenly distributed, with a left-skewed concentration in the 72–90% range. That suggests widespread but not universal screening practices.

Outlier Detection & Investigation

With the a z-score method, we were able to flag any state-year observation with a z-score greater than 2.5, to find outliers. This approach highlighted several instances in 2021 that are likely tied to pandemic disruptions in preventive care. California, Washington D.C., and Washington state showed unusually low rates of **cholesterol checks** (~72–74% vs. an 83.8% mean), while West Coast states (CA, NM, OR, WA) also reported lower-than-average rates for **routine checkups**, a pattern consistent aligning with the COVID-19 disruption idea. In a different trend, the District of Columbia consistently was a high outlier for **video calling** from 2021–2023 (85–86% vs. a 60% mean), which could reflect its urban, professional, and tech-heavy population. We retained all of these outliers since they represent genuine state-level variations due to policy, demographics, or pandemic response rather than data quality issues.



Exploratory Data Analysis - State Patterns & Correlations

Scatterplots were created to test direct relationships between digital infrastructure and healthcare outcomes. Overall, **linear patterns are weak or absent**:

1. **Internet access vs routine checkups:** No clear trend; states with higher internet penetration still showed wide variation in checkup rates.
2. **PC/tablet ownership vs personal doctor access:** Points are widely dispersed, suggesting other factors (insurance, healthcare supply, Medicaid expansion) dominate this relationship.
3. **Video calling vs flu shots:** Highly scattered; telehealth capability does not align with preventive vaccination uptake.
4. **Overall digital access vs overall healthcare engagement:** A slight upward slope is visible, but the scatter is substantial, indicating only a weak overall link.

Key Insight: Digital infrastructure by itself does **not directly predict healthcare engagement**. The weak associations point to the importance of **confounders** such as socioeconomic status, demographics, and rural/urban divides. These are variables that are highly impacting on healthcare engagement. This reinforces why we moved beyond scatterplots to **correlation and regression models**, to quantify which digital dimensions (like email use vs video calling) matter most, and how strongly.



Correlation & Regression Analysis

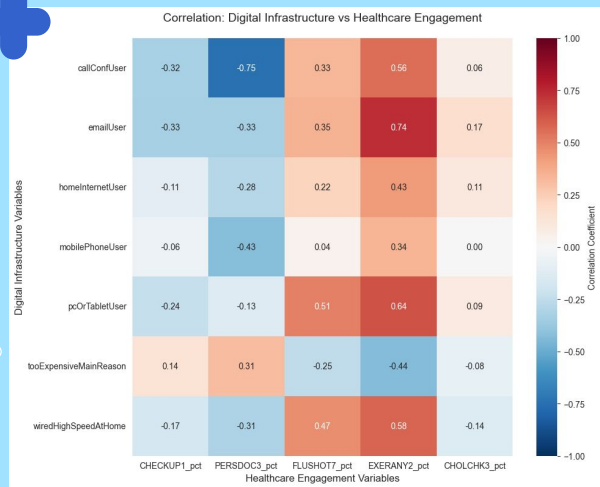
Pearson Correlation Coefficients (n=146 state-years)

Strongest Positive Correlations:

- Email Use ↔ Exercise: **$r = 0.740$**
- PC/Tablet ↔ Exercise: **$r = 0.639$**
- Broadband ↔ Exercise: **$r = 0.580$**
- Video Calling ↔ Exercise: **$r = 0.561$**
- PC/Tablet ↔ Flu Shots: **$r = 0.507$**
- Broadband ↔ Flu Shots: **$r = 0.468$**

Strongest Negative Correlations:

- Video Calling ↔ Personal Doctor: **$r = -0.747$**
- Mobile Phone ↔ Personal Doctor: **$r = -0.429$**
- Email ↔ Personal Doctor: **$r = -0.429$**
- Email ↔ Checkups: **$r = -0.329$**



High vs Low Digital States (T-Tests)

Grouped by composite digital score (top 25% vs bottom 25%, n=37 each)

Healthcare Variable	High Digital	Low Digital	Difference	p-value
Personal Doctor	67.0%	84.0%	-17.0	p<0.001***
Exercise	78.9%	71.8%	+7.1	p<0.001***
Flu Shots	47.6%	41.4%	+6.2	p<0.001***
Checkups	85.8%	87.9%	-2.0	p=0.003**
Cholesterol	13.1%	11.9%	+1.1	p=0.258 (ns)

High digital infrastructure states show a **17 percentage point deficit** in personal doctor access compared to low digital states, despite the advantages in exercise (+7.1 points) and flu shots (+6.2 points). Digital tools may be diminishing traditional healthcare practices rather than enhancing them. States with higher video calling, mobile phone use, and email show significantly lower rates of having a personal healthcare provider. This could suggest a shift from primary care relationships toward telehealth, urgent care, or specialist direct access. **But correlation ≠ causation.** Digital infrastructure likely are proxies for urbanization, education, income, and demographics, all confounding factors not controlled for in this analysis.

Pattern: Digital infrastructure correlates positively with lifestyle behaviors (exercise) but negatively with traditional care access (personal doctor).

Linear Regression Models

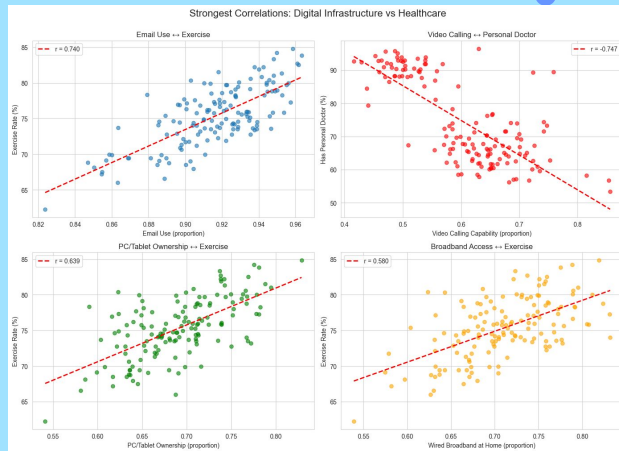
All models $p < 0.001$ (very significant)

Model 1: Exercise = $-31.01 + 116.08 \times \text{Email}$, **$R^2 = 0.548$** (explains 54.8% of variance), 10-point increase in email → 11.6-point increase in exercise

Model 2: PersonalDoc = $137.48 - 104.42 \times \text{Video}$, **$R^2 = 0.558$** (strongest model), 10-point increase in video calling → 10.4-point decrease in personal doctor access

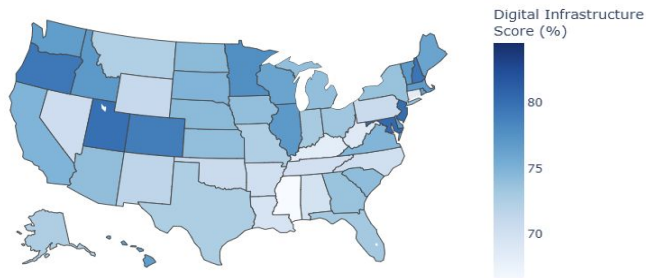
Model 3: FluShots = $13.07 + 43.75 \times \text{Broadband}$, **$R^2 = 0.219$** , 10-point increase in broadband → 4.4-point increase in flu shots

Model 4: Exercise = $39.55 + 51.74 \times \text{PCTablet}$, **$R^2 = 0.408$** ,

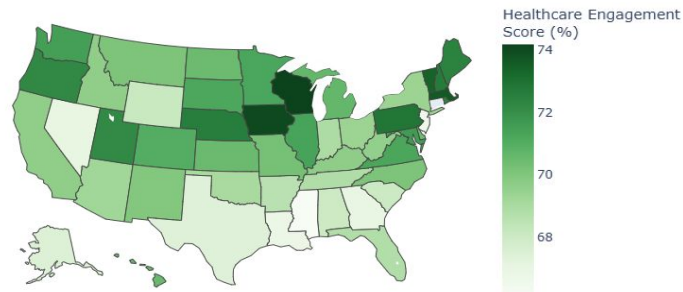


Geographic Visualizations & Temporal Trends

State-Level Digital Infrastructure (2019-2023 Average)



State-Level Preventive Healthcare Utilization (2019-2023 Average)



Geographic Patterns - Digital Infrastructure

Highest Digital Infrastructure (72-75%):

- **Northeast:** NH, MA, CT, RI, VT
- **Mid-Atlantic:** DC, MD, DE
- **Mountain West:** CO, UT
- **Pacific Northwest:** WA

Lowest Digital Infrastructure (57-62%):

- **Deep South:** MS, LA, AL, AR
- **Appalachia:** WV, KY
- **Southwest:** NM

Pattern: Clear urban-rural divide. Metropolitan regions and coastal states show stronger digital infrastructure adoption.

Geographic Patterns - Healthcare Engagement

Highest Healthcare Engagement (63-65%):

- **Upper Midwest:** WI (#1), MN
- **Northeast:** VT, NH, ME
- **Pacific Northwest:** WA

Lowest Healthcare Engagement (52-54%)

- **Deep South:** MS, LA, GA, SC
- **Southwest:** NV, NM

Pattern: Northern and western states rank higher; southern states rank consistently lower. But there are outliers (i.e Wisconsin ranks **#1 healthcare** but only moderate digital).

Temporal Trends (2019 → 2021 → 2023)

We analyzed how metrics changed over our 5-year study period across the COVID-19 pandemic:

Digital Infrastructure Trends:

- **Email use:** 89% → 92% → 94% (steady growth)
- **Video calling:** 59% → 71% → 78% (rapid pandemic adoption)
- **Broadband:** 76% → 79% → 81% (moderate growth)
- **Trend:** Digital infrastructure **consistently improved** across all states

Healthcare Engagement Trends:

- **Exercise:** 75% → 74% → 76% (minimal change)
- **Flu shots:** 44% → 43% → 45% (stable)
- **Checkups:** 87% → 86% → 88% (stable)
- **Personal doctor:** 81% → 72% → 61% (**-20 percentage points!**)

Digital infrastructure rose steadily, but healthcare engagement did not move in parallel. Preventive care behaviors held steady, while access to personal doctors collapsed, showing that digital growth is reshaping how people engage with care, not necessarily improving traditional access.

Discussion

Summary of Key Findings

Our analysis explored whether stronger digital infrastructure correlates with greater healthcare engagement across U.S. states. The findings reveal a **complex and contradictory relationship**: digital access supports some forms of engagement while weakening others. States with higher digital use (particularly email and broadband) show **significantly higher rates of preventive behaviors** like exercise ($r = 0.74$) and flu shots ($r = 0.47$). However, these same states report **dramatically lower access to personal doctors** (17 percentage point gap between high and low digital states) and fewer routine checkups. This suggests that digital tools may be **displacing rather than expanding traditional care relationships**. The temporal analysis revealed a striking pattern: from 2019 to 2023, personal doctor access **declined 20 percentage points** ($81\% \rightarrow 61\%$) while digital infrastructure steadily improved. Regression models show digital variables explaining up to **56% of healthcare behavior variance**, demonstrating their statistical relevance. Geographic patterns showed regional clustering: Northeast and Pacific Northwest states rank high on both metrics, while Southern states rank consistently lower. However, the overlap is imperfect, suggesting other regional factors at play.

Limitations

While the relationships are statistically significant, they should be interpreted with caution to do several limitations.

First, **correlation does NOT equal causation**. We can't determine whether digital infrastructure causes these patterns or whether both are driven by other factors.

Second, **demographic controls were not included** in our state-level analysis. Variables like income, education, age, and urban/rural status are confounding variables that could affect the observed relationships. Urban states tend to have both better internet access and distinct healthcare patterns, making it impossible to isolate digital infrastructure's independent effect without controlling for these confounding factors.

Third, **state-level aggregation** hides individual experiences and within-state variation. Even in states with 70% internet access, the remaining 30% without access are overlooked in the analysis.

Fourth, the **COVID-19 pandemic (2019-2023)** disrupted healthcare systems, making it difficult to separate pandemic effects from underlying digital infrastructure relationships. The sharp decline in personal doctor access may reflect doctor shortages and system collapse rather than digital impacts.

Fifth, our models are **simple linear regressions** that may not capture nonlinear dynamics or complex interactions between variables.

Outlier Detection & Investigation

Future work should incorporate more individual microdata instead of joining by states so we can control for socioeconomic confounders (income, education, age, race). This would better isolate digital infrastructure effects. **Demographic controls** should be included to understand how much of the observed relationship is truly about digital access rather than underlying population characteristics. A bigger focus in area comparison like county-level or urban vs. rural) is needed to understand inter-state disparities. **Longitudinal panel data** tracking the same areas over longer time periods (including pre-pandemic time) would help separate COVID-19 disruptions from possible trends. One variable to explicitly measure in the future is **telehealth adoption** and we should investigate whether digital tools are genuinely improving access or just shifting care from primary care to urgent care, telehealth, or direct specialist access without improving outcomes. Even adjusting the methodology of this study, such as combining quantitative analysis with qualitative interviews, could help reveal the actual factors by which digital infrastructure influences healthcare engagement. These future directions can advance research on digital access, its use, and its potential to enable more positive healthcare-related behaviors. It remains crucial to continue isolating key factors and controlling for confounding variables in order to accurately identify correlations between digital access and healthcare behaviors.



References

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